

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

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(An Autonomous Institute, with NAAC 'A' Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of CSE

'A place to learn; A chance to grow'

2025-2026 (Odd Semester)

SEMESTER EXAMINATION

Course: B.tech in CSE

Subject: Blockchain Technology

Max Marks: 20

Date: 03/10/2025

Year/Semester: 5th Semester (3rd Year)

Subject Code: CS5E001A

Duration: - 1 Hr.

Instructions to the Students:

1. All the questions carry marks as indicated.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

CO 1: Students will be able to Understand emerging abstract models for Blockchain Technology.

CO 2: Students will be able to Identify major research challenges and technical gaps existing between theory and practice in the crypto currency domain

CO 3: Students will be able to provide conceptual understanding of the function of Blockchain as a method of securing distributed ledgers, how consensus on their contents is achieved, and the new applications that they enable

CO 4: Student will able to Apply hyper ledger Fabric and Ethereum platform to implement the Blockchain Application

CO 5: Student will able to design applications based on blockchain technology for E-Governance, Land Registration, Medical Information Systems, and others

SECTION A

1. All Questions are compulsory.
2. Answer in one word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	List out the four names of the consensus mechanism	L1/CO4	1
Q.2	Give the name of the nodes used in paxos algorithm	L1/CO3	1
Q.3	Define power of burn in blockchain technology.	L1/CO3	1
Q.4	State advantages of Ethereum Technology (any two)	L1/CO4	1

SECTION B

1. All Questions are compulsory.
2. Solve answers in brief (two, three Sentence).

Q.5	Define DAO in ethereum Technology.	L1/CO4	2
Q.6	State about smart contracts in Ethereum Technology	L1/CO4	2
Q.7	Discuss the types of consensus mechanism (any two)	L2/CO3	2

Q 8	Distinguish between POS and POW.	L2/CO3	2
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SECTION C

1. Solve any two questions.

Q 9	Describe byzantine algorithm in block-chain technology with diagram	L6/CO3	4
Q 10	Explain PAXOS algorithm in Blockchain Technology	L5/CO3	4
Q 11	Illustrate the following-1.Ethereum 2.Smart Contract	L4/CO4	4